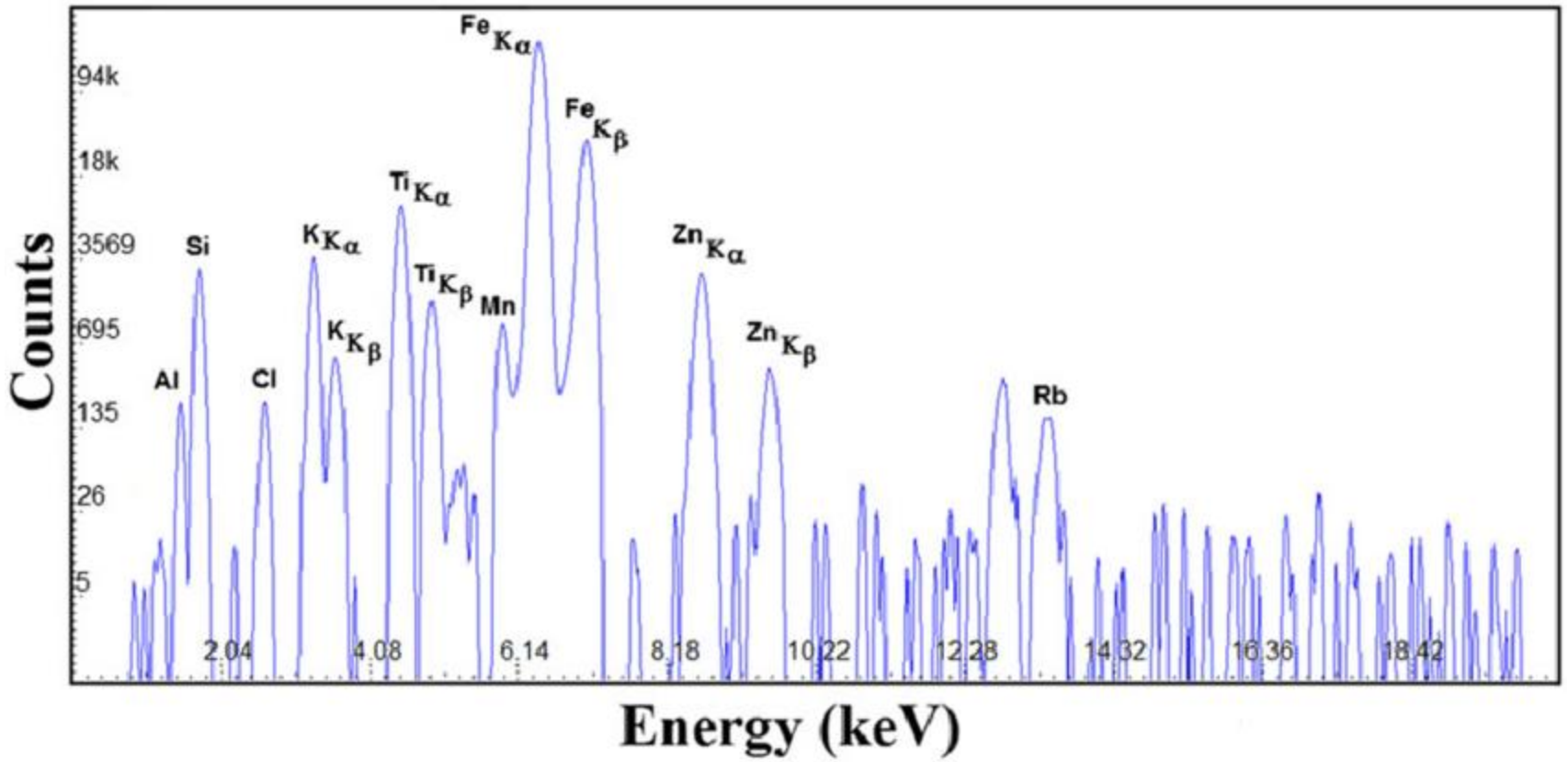
XRF


Counts
94k
18k
3569
695
135
26
5
Al
Si
Cl
K Kα
K Kβ
Ti Kα
Ti Kβ
Mn
Fe Kα
Fe Kβ
Zn Kα
Zn Kβ
Rb
2.04
4.08
6.14
8.18
10.22
12.28
14.32
16.36
18.42
Energy (keV)

X-Ray

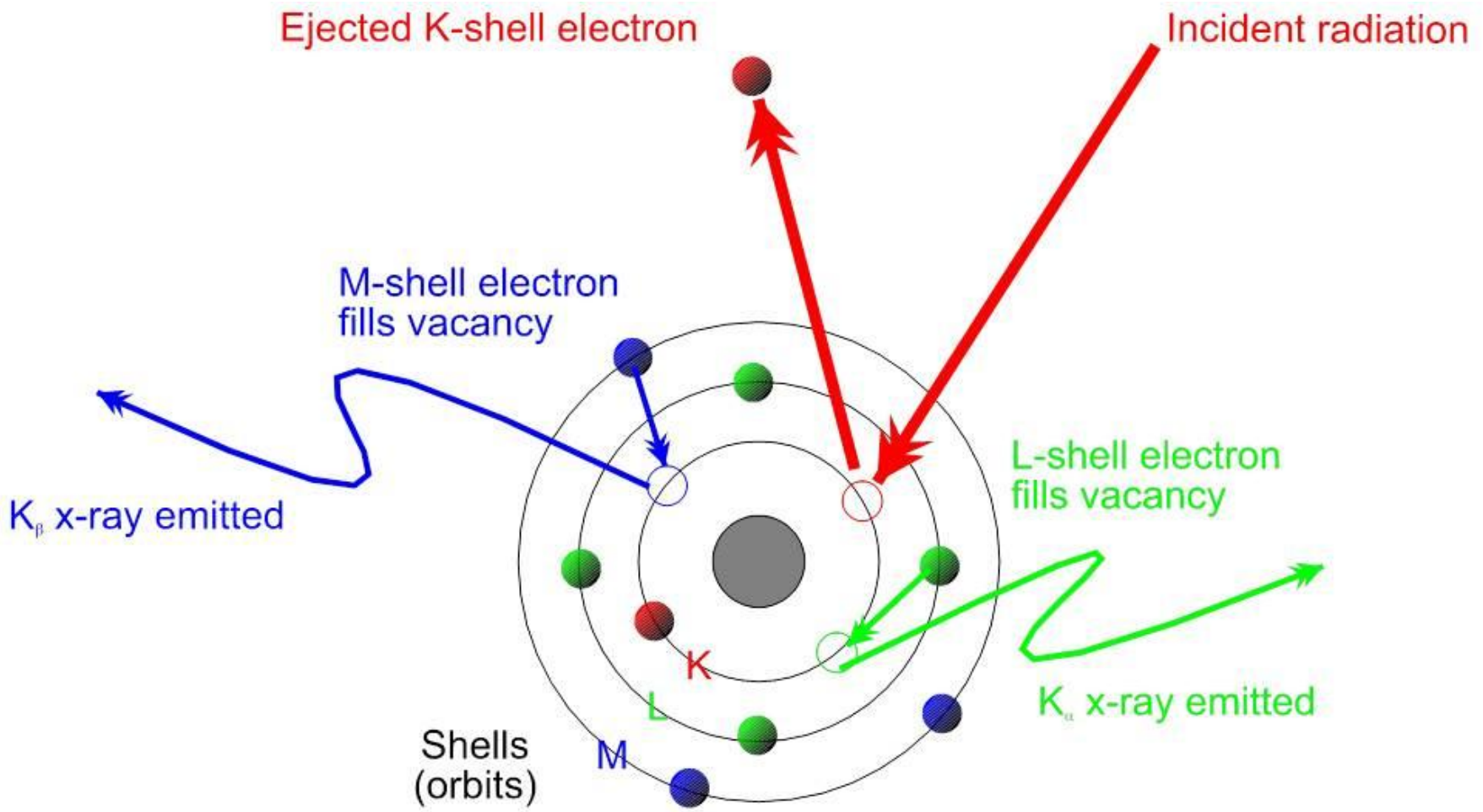
- X-rays (Wilhelm Röntgen, 1895) form part of the electromagnetic spectrum, and are characterized by energies lying between ultra-violet and gamma radiation.

- Wavelengths are typically in the range 0.01 to 10 nm, which is equivalent to energies of 125 keV to 0.125 keV.

- Widely used including for medical imaging, baggage screening, and elemental and structural analytical techniques.

X-ray interactions with Objects

- Absorption, scattering, transmitted.
- These interactions depends on the sample thickness, density and composition, and the X-ray energy.
- When a primary x-ray excitation source from an x-ray tube or a radioactive source strikes a sample, the x-ray can either be absorbed by the atom or scattered through the material.
- The process in which an x-ray is absorbed by the atom by transferring all of its energy to an inner most electron is called the “photo-electric effect.”
- During this process, if the primary x-ray had sufficient energy, electrons are ejected from the inner shells, creating vacancies. These vacancies present an unstable condition for the atom.


Ejected K-shell electron
Incident radiation
M-shell electron
fills vacancy
K_β x-ray emitted
L-shell electron
fills vacancy
K_α x-ray emitted
K
L
M
Shells
(orbits)

XRF

- As the atom returns to its stable condition, electrons from the outer shells are transferred to the inner-shells and in the process giving off a characteristic x-ray whose energy is the difference between the two binding energies of the corresponding shells.

- The emitted x-rays produced from this process are called "X-ray Fluorescence," or XRF.

- The process of detecting and analyzing the emitted x-rays is called "X-ray Fluorescence Analysis."

- The energy of the emitted X-ray for a particular element is independent of the chemistry of the material.
  - For example, a calcium peak obtained from $CaCO_3$, CaO and $CaCl_2$ will be in exactly the same spectral position for all three materials.

XRF Principle

- X-ray Fluorescence (XRF) can be considered in a simple three step process occurring at the atomic level:
  1. An incoming X-ray knocks out an electron from one of the orbitals surrounding the nucleus within an atom of the material.
  2. A hole is produced in the orbital, resulting in a high energy, unstable configuration for the atom.
  3. To restore equilibrium, an electron from a higher energy, outer orbital falls into the hole. Since this is a lower energy position, the excess energy is emitted in the form of a fluorescent X-ray.

- **The energy difference between the expelled and replacement electrons is characteristic of the element atom** in which the fluorescence process is occurring – thus, the energy of the emitted fluorescent X-ray is directly linked to a specific element being analyzed.

- It is this key feature which makes XRF such a fast analytical tool for elemental composition.

Transitions

- Since most atoms comprise a number of electron orbitals (eg, K shell, L shell, M shell) a number of possible fluorescent transitions are possible.
- For example, interaction of X-rays with an atom with K, L and M shells could result in a hole forming in the K shell, which is then filled by an electron from the L shell or from the M shell.
- The characteristic x-rays are labeled as K, L, M or N to denote the shells they originated from.
- Another designation alpha(α), beta(β) or gamma(γ) is made to mark the x-rays that originated from the transitions of electrons from higher shells.
- Hence, a Kα x-ray is produced from a transition of an electron from the L to the K shell, and a Kβ X-ray is produced from a transition of an electron from the M to a K shell, etc.
- Thus, for a single element, a number of XRF peaks are possible, and typically these will all be present in the spectrum, with varying intensities. They form a characteristic fingerprint for a specific element.

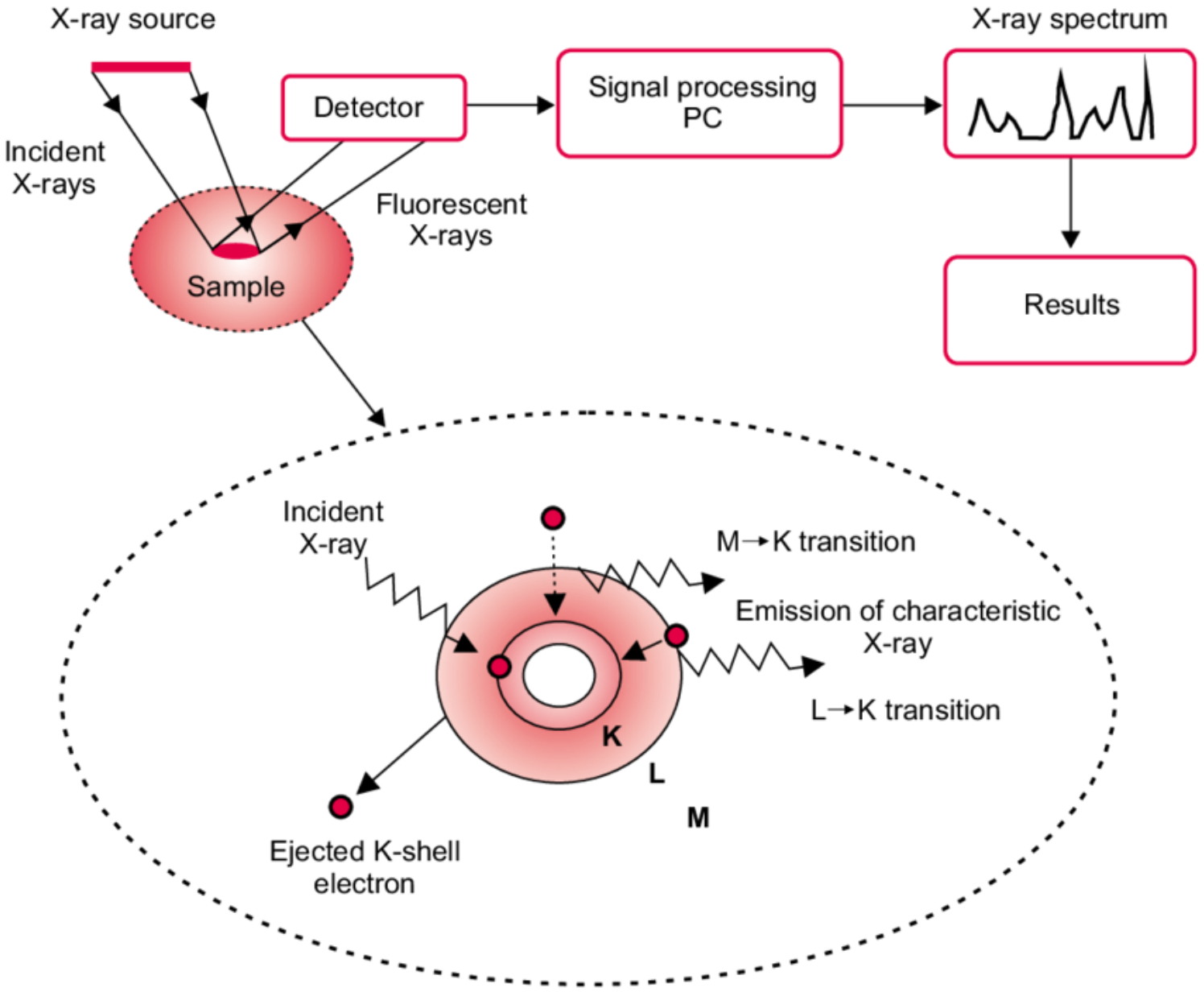
The Work Flow

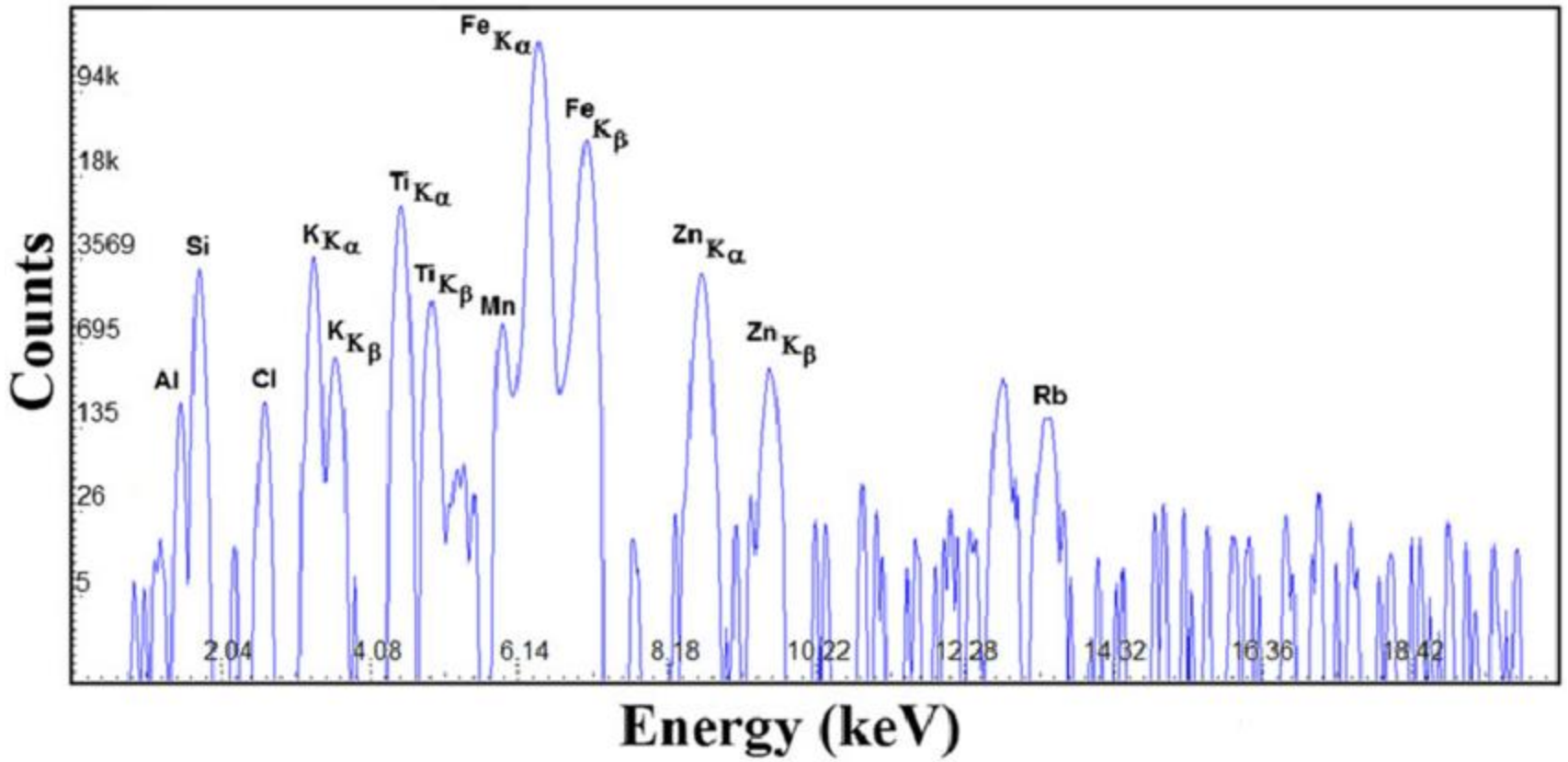
1. A solid or a liquid sample is irradiated with high energy X-rays from a controlled X-ray tube.

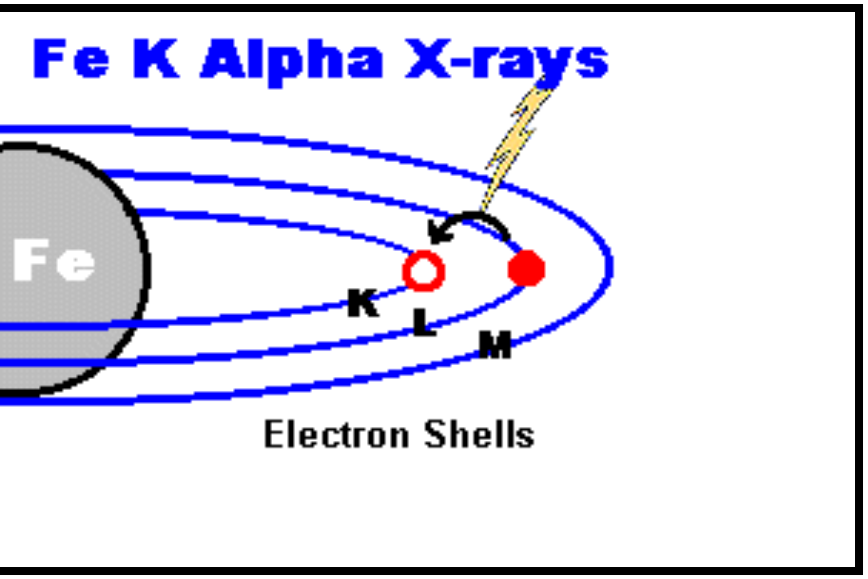
2. When an atom in the sample is struck with an X-ray of sufficient energy (greater than the atom's K or L shell binding energy), an electron from one of the atom's inner orbital shells is dislodged.

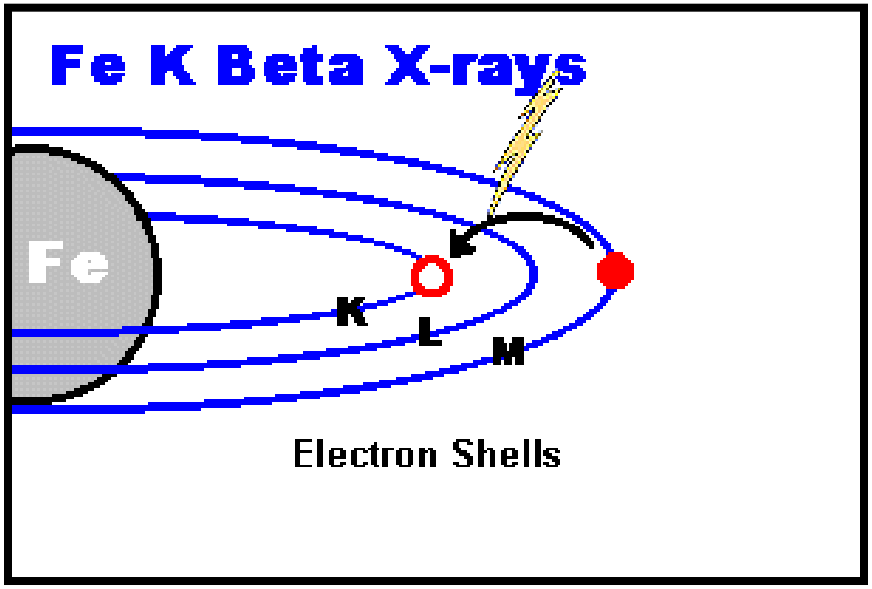
3. The atom regains stability, filling the vacancy left in the inner orbital shell with an electron from one of the atom's higher energy orbital shells.

4. The electron drops to the lower energy state by releasing a fluorescent X-ray. The energy of this X-ray is equal to the specific difference in energy between two quantum states of the electron. The measurement of this energy is the basis of XRF analysis


X-ray source
X-ray spectrum
Detector
Signal processing
PC
Incident
X-rays
Fluorescent
X-rays
Sample
Results
Incident
X-ray
M→K transition
Emission of characteristic
X-ray
L→K transition
K
L
M
Ejected K-shell
electron

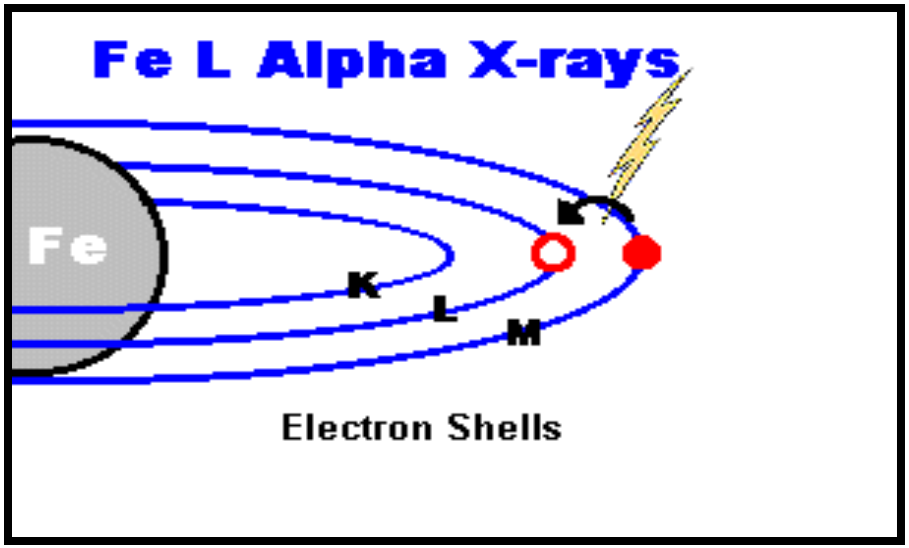

Counts
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Energy (keV)





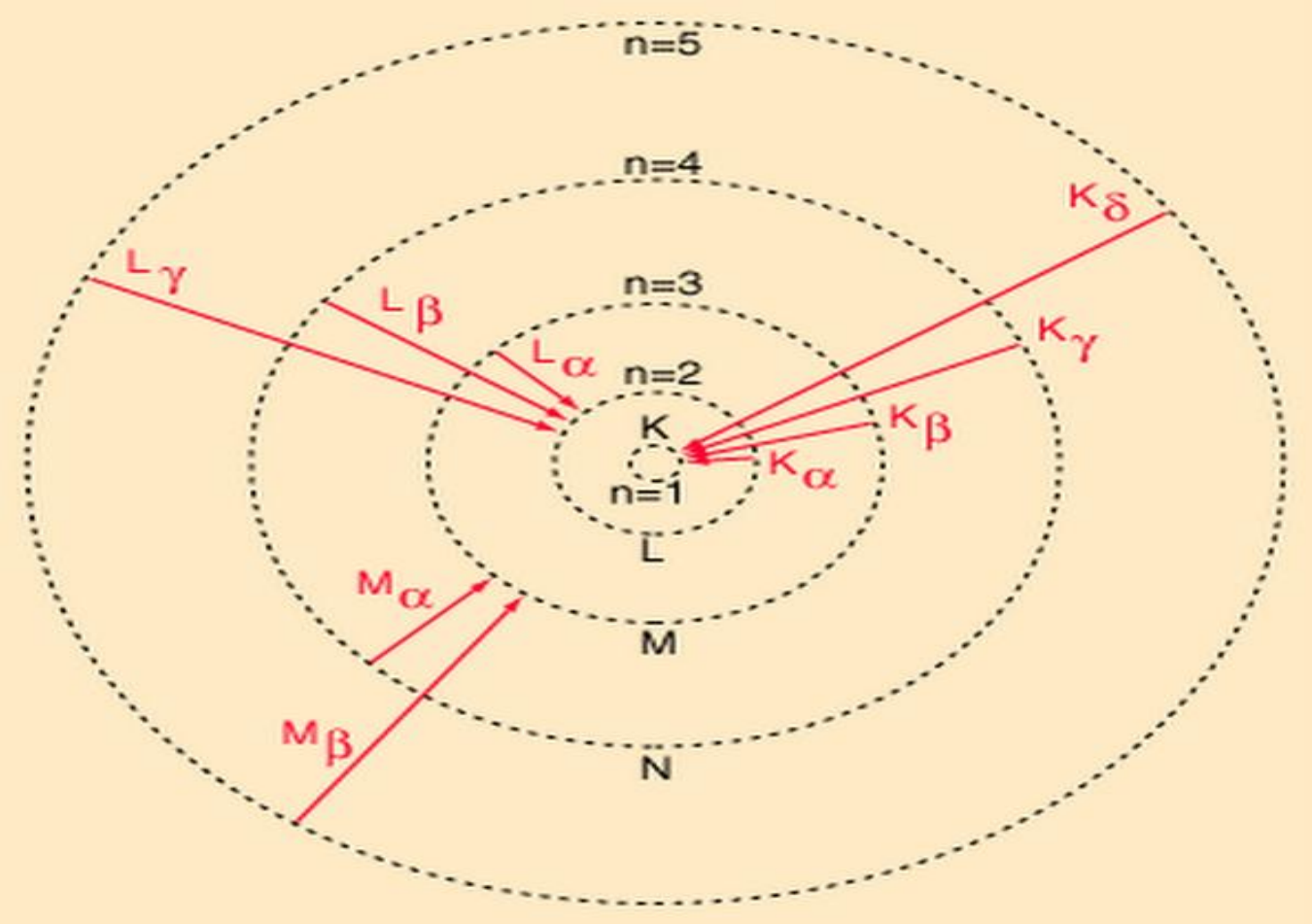
•When an electron from a K-shell is replaced by one from the next closest shell (L), it is designated as a **Kα event**

•When an electron from a K-shell is replaced by one from the second closest shell (M), it is designated as a **Kβ event**



Lα - When an electron from a L-shell is replaced by one from the next closest shell (M).

The K shell will never donate its electron as this would require an increase in energy, not a drop.



•Certain events such as **Mα, Lβ, and Kγ** are only possible in atoms of sufficient atomic weight.

•**Each element has a family of characteristic X-rays associated with it**

Table of X-ray Emission and Absorption Edge Energies (keV)

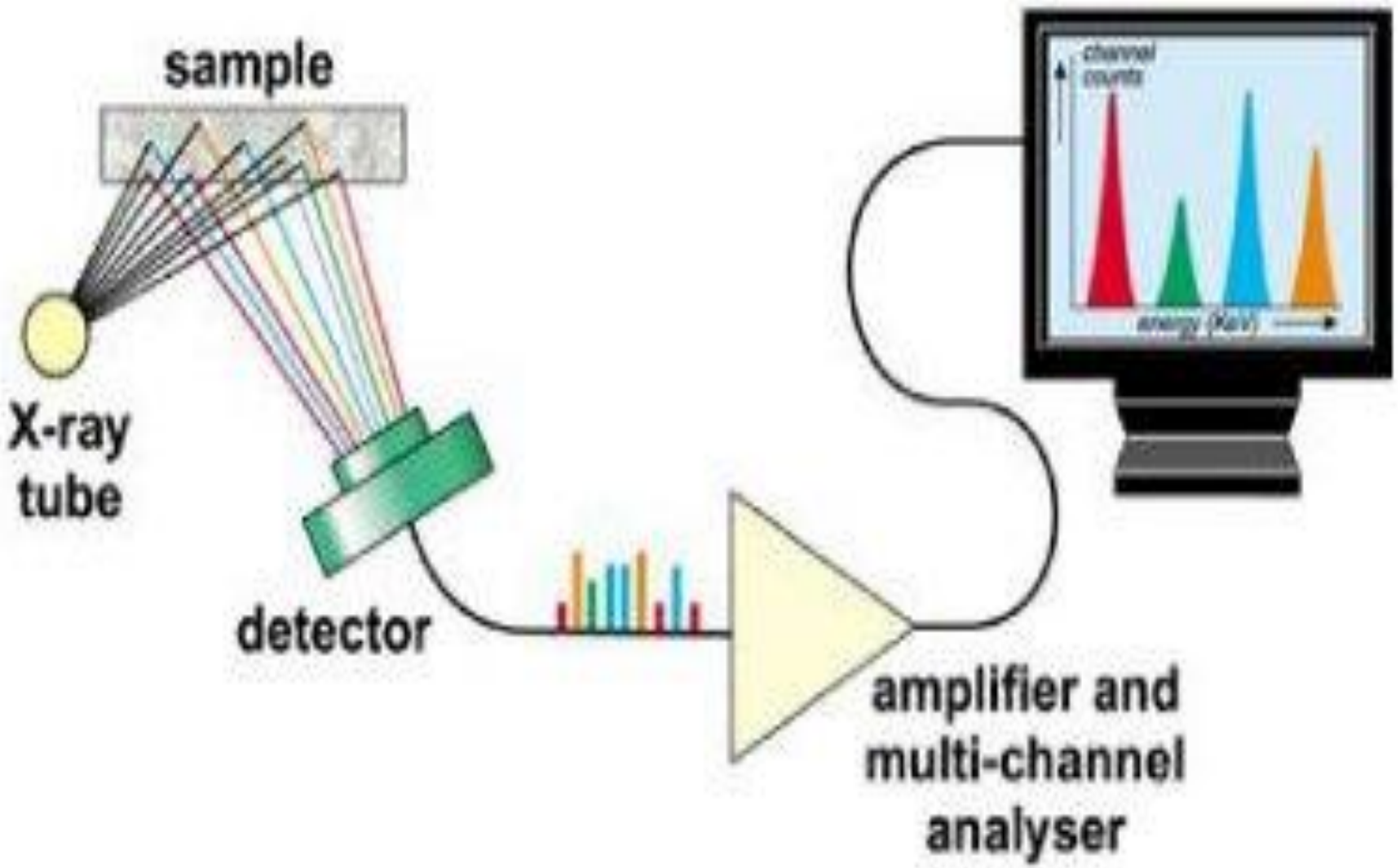
atomic number	Kα	Kβ	$K_{(ab)}$	Lα	$L_{\beta 1}$	$L_{\beta 2}$	$L_{\gamma 1}$	$L_{III(ab)}$	$L_{II(ab)}$	$M_{V(ab)}$	Mα
1 H			0.014								
2 He			0.025								
3 Li			0.056					0.003	0.003		
4 Be			0.112					0.002	0.002		
5 B			0.186					0.003	0.003		
6 C	0.277		0.282					0.002	0.002		
7 N	0.392		0.400					0.008	0.008		
8 O	0.525		0.531					0.009	0.009		
9 F	0.677		0.692								
10 Ne	0.848		0.874					0.019	0.019		
11 Na	1.041	1.067	1.070					0.031	0.031		
12 Mg	1.253	1.295	1.303					0.050	0.050		
13 Al	1.486	1.553	1.559					0.073	0.073		
14 Si	1.739	1.829	1.842					0.102	0.103		
15 P	2.013	2.136	2.142					0.128	0.129		
16 S	2.307	2.464	2.470					0.163	0.164		
17 Cl	2.621	2.815	2.820					0.197	0.199		
18 Ar	2.957	3.190	3.200					0.224	0.246		
19 K	3.312	3.589	3.609					0.295	0.298		
20 Ca	3.690	4.012	4.038	0.341	0.345		0.350	0.346	0.350		
21 Sc	4.088	4.460	4.496	0.396	0.400		0.407	0.403	0.411		
22 Ti	4.508	4.931	4.964	0.452	0.458		0.460	0.454	0.460		
23 V	4.949	5.426	5.464	0.511	0.519		0.520	0.513	0.519		
24 Cr	5.411	5.924	5.987	0.573	0.583		0.583	0.574	0.581		
25 Mn	5.894	6.489	6.537	0.637	0.649		0.652	0.641	0.650		
26 Fe	6.398	7.057	7.111	0.705	0.718		0.721	0.709	0.720		
27 Co	6.924	7.648	7.709	0.776	0.791		0.794	0.779	0.794		
28 Ni	7.471	8.263	8.331	0.851	0.869			0.854	0.871		
29 Cu	8.040	8.904	8.981	0.930	0.950			0.933	0.953		
30 Zn	8.630	9.570	9.661	1.012	1.034			1.022	1.045		

Instrumentation

- The key components of a typical XRF spectrometer are:
  1. Source of X-rays used to irradiate the sample.
  2. Sample.
  3. Detection of the emitted fluorescent X-rays.

- The resulting XRF spectrum shows intensity of X-rays (usually in counts per second) as a function of energy (usually in eV).

- There are two main types of XRF spectroscopy. Energy Dispersive XRF (EDXRF) and Wavelength Dispersive XRF (WDXRF), which differ primarily in the way the fluorescent X-rays are detected and analyzed.


sample
X-ray tube
detector
amplifier and multi-channel analyser
channel counts
energy (KeV)

Types of XRF

- **Energy Dispersive XRF**
  - An energy dispersive detection system directly measures the different energies of the emitted X-rays from the sample. By counting and plotting the relative numbers of X-rays at each energy an XRF spectrum is generated.

- **Wavelength Dispersive XRF**
  - A wavelength dispersive detection system physically separates the X-rays according to their wavelengths.
  - The X-rays are directed to a crystal, which diffracts the X-rays in different directions according to their wavelengths (energies).

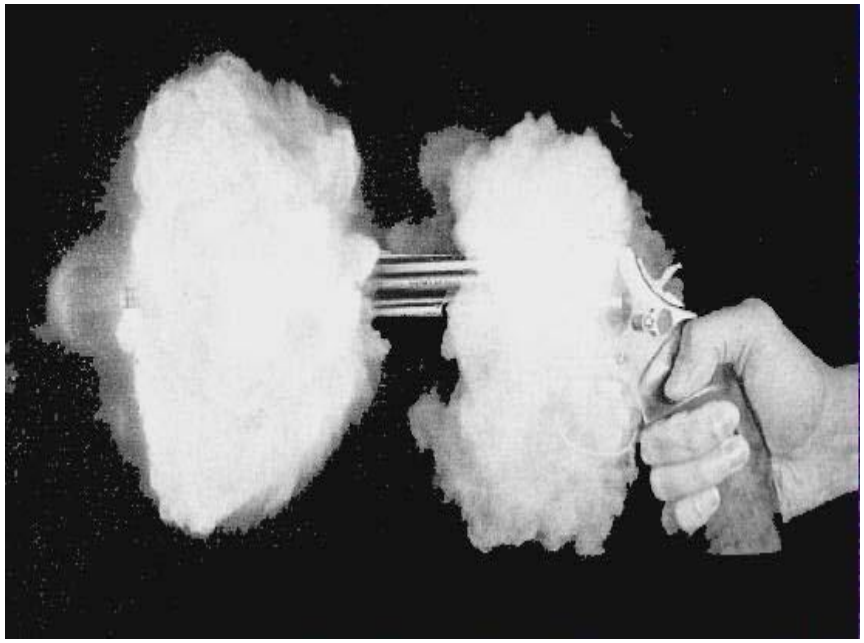
APPLICATIONS

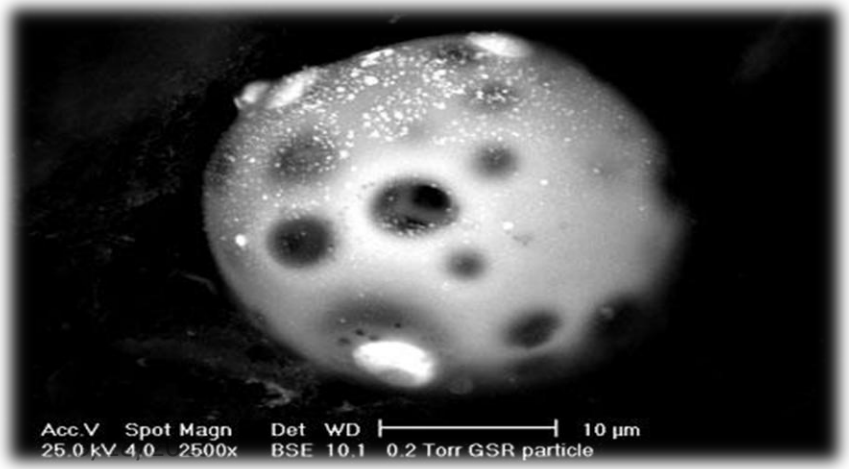
- X-Ray fluorescence is used in a wide range of applications-
  - Research in igneous, sedimentary, and metamorphic petrology
  - Soil surveys
  - Mining (e.g., measuring the grade of ore)
  - Cement production
  - Ceramic and glass manufacturing
  - Metallurgy (e.g., quality control)
  - Environmental studies (e.g., analyses of particulate matter on air filters)
  - Petroleum industry (e.g., sulfur content of crude oils and petroleum products)
  - Field analysis in geological and environmental studies (using portable, hand-held XRF spectrometers)

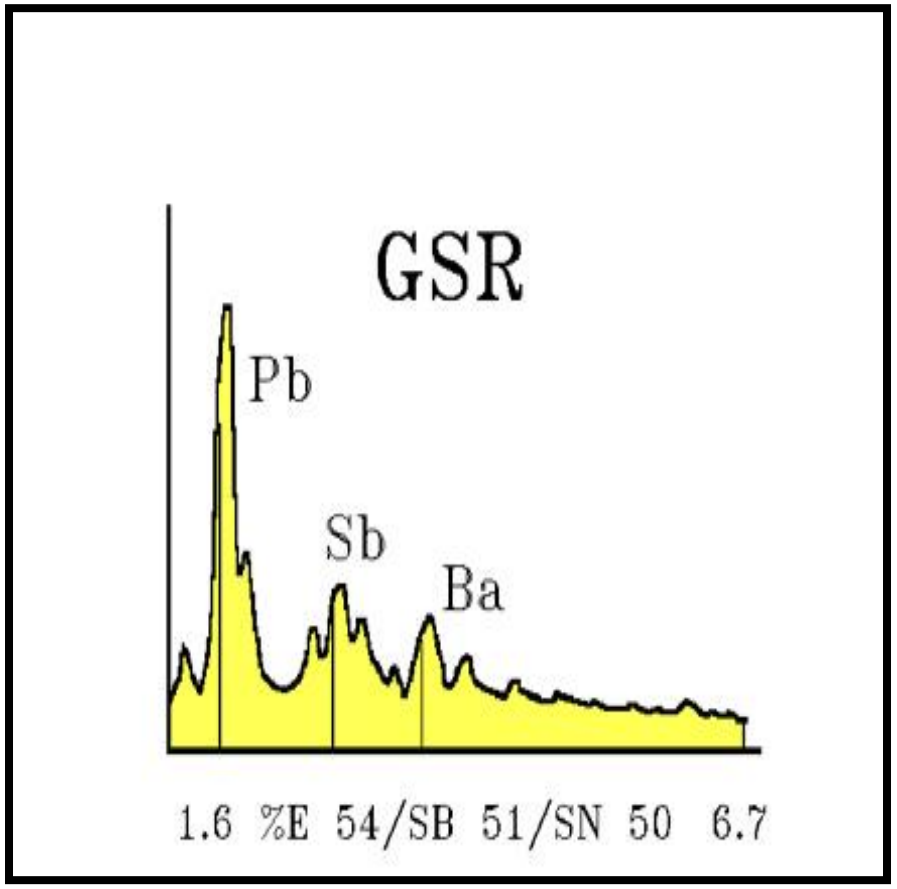
Gunshot Residue (GSR) Analysis

- GSR is often found on criminals and also on victims if shot at close range.
- Particles are very characteristic, therefore presence of these particles forms evidence of firing a gun.
- Particles normally consist of Pb (lead), Sb (antimony) and Ba (barium).
- The proportion of elements present in GSR differ slightly and databases of GSR from different manufacturers can be used to identify what ammunition was used in a crime.
- New ammunition: environmentally friendly (no Sb).

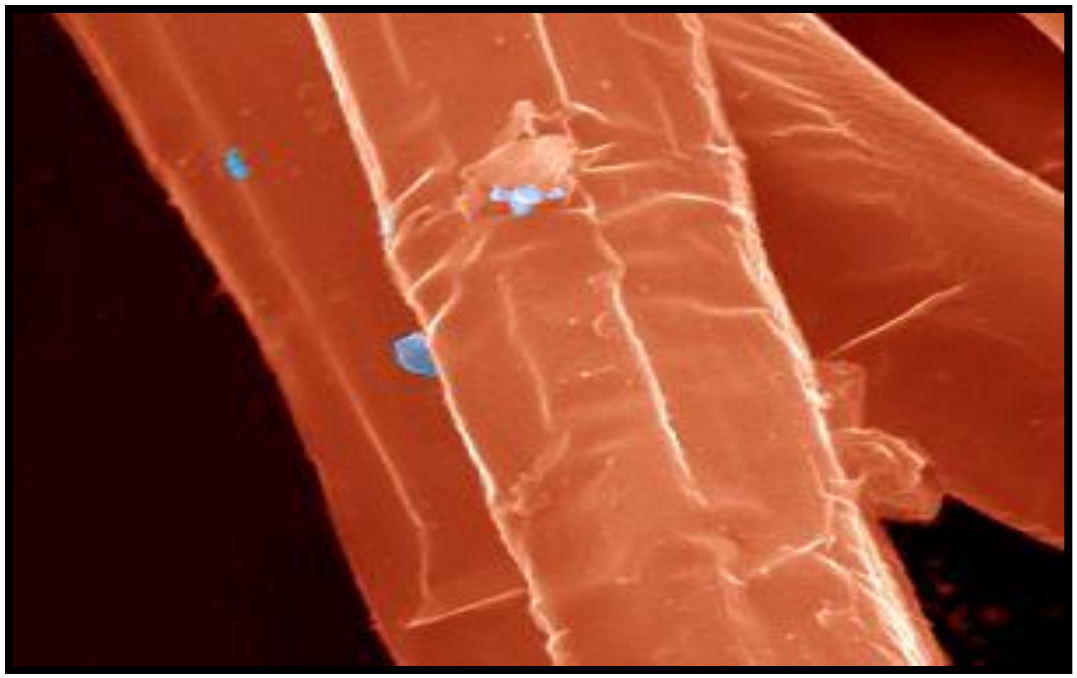
Gunshot Residue (GSR) Analysis



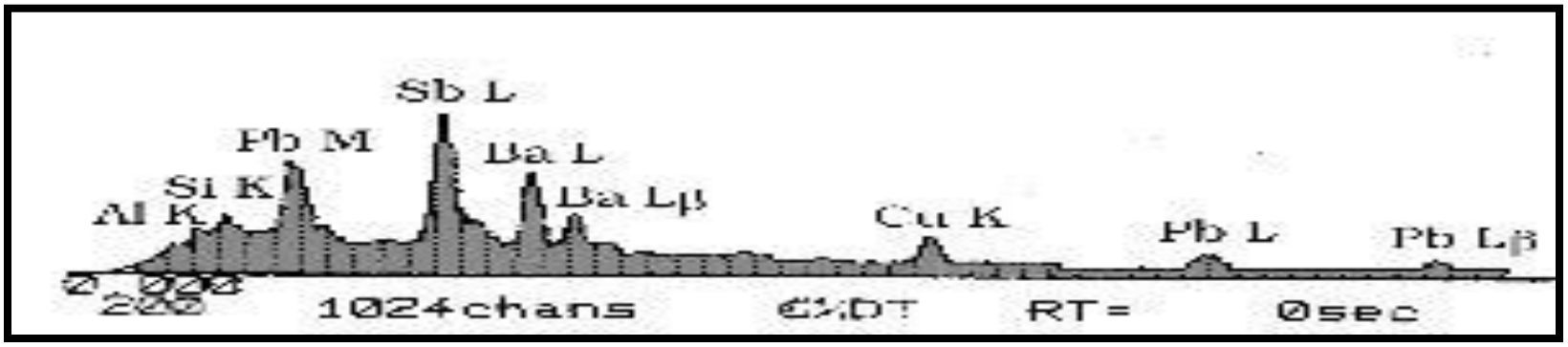




GSR Analysis on cloth fibre



•Bullet fragments (blue) can be identified on cloth fibres and distinguished from other metal pieces by their elemental composition



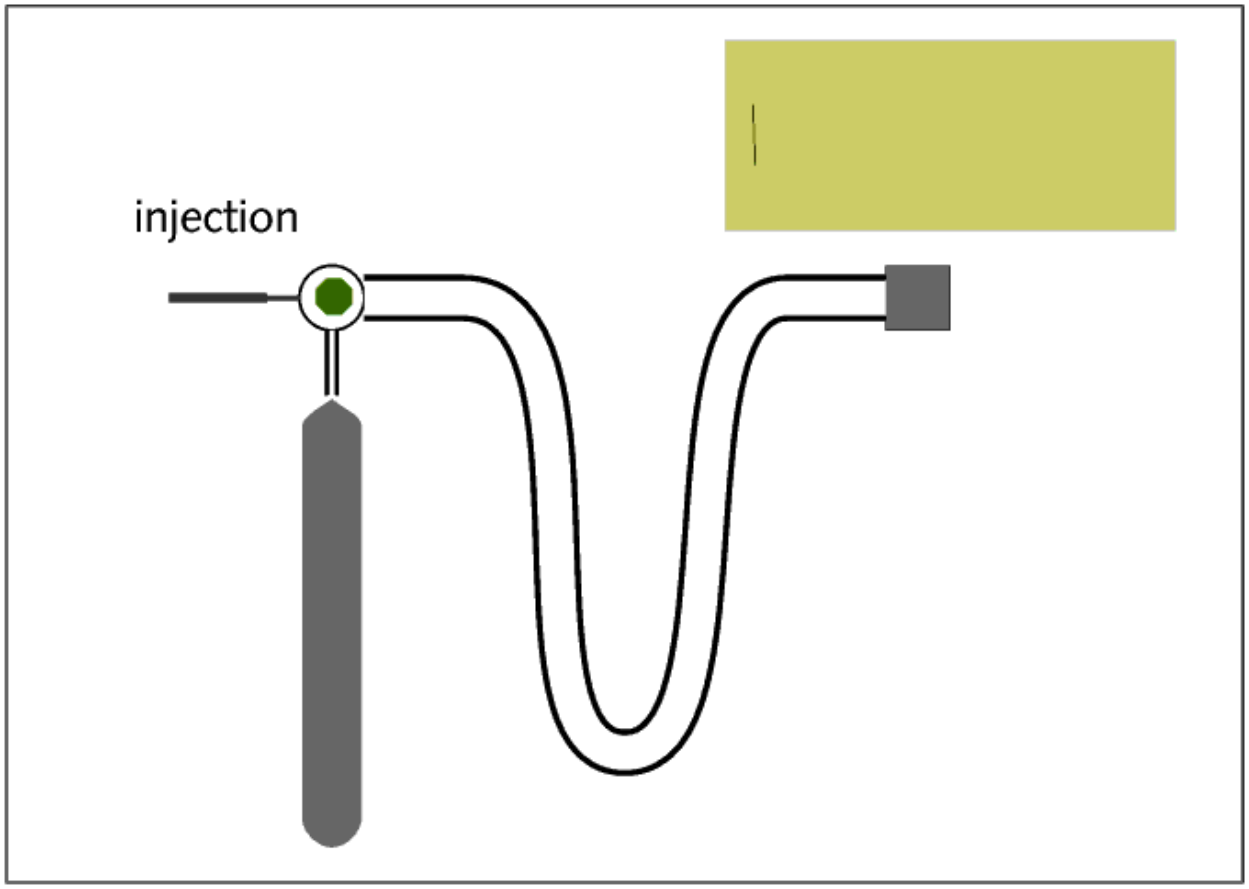
Is XRF safe?

- During the analysis, the analyzer emits a directed radiation beam when the tube is energized.

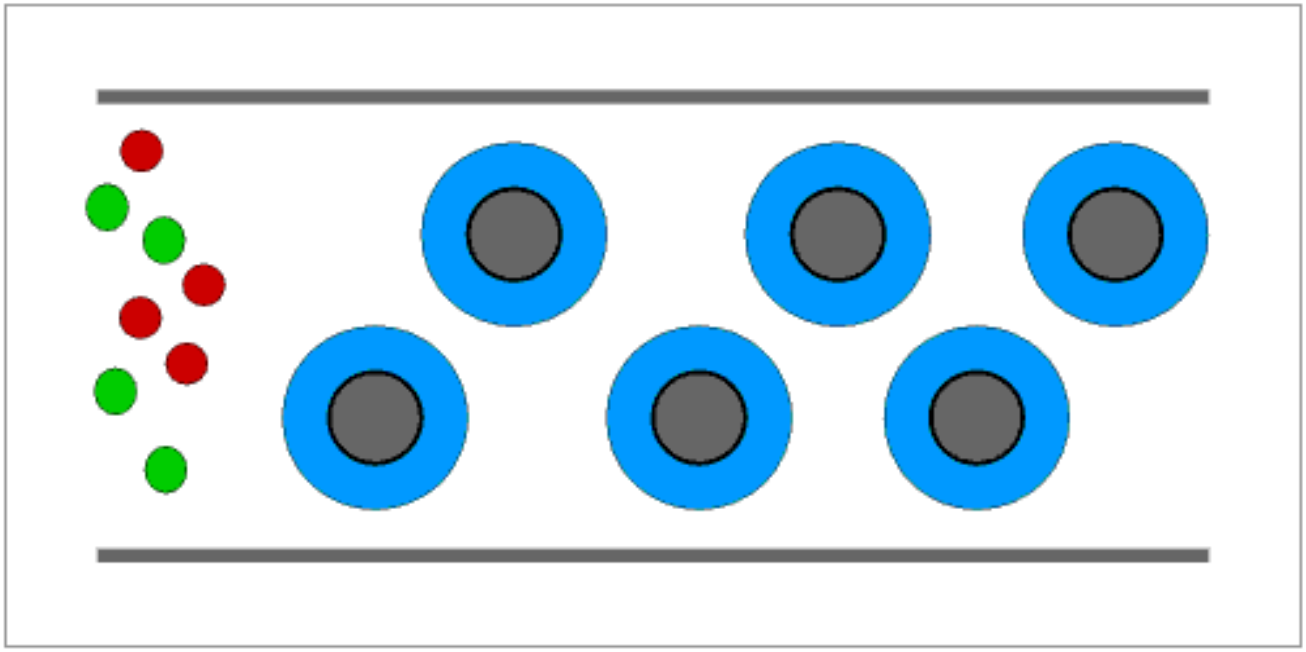
- Reasonable effort should be made to maintain exposures to radiation as far below dose limits (ALARA; As Low as Reasonably Achievable principle).

- Three factors will help minimize your radiation exposure: time, distance, and shielding.
  1. Provide radiation safety training to operators
  2. Never aim the device at yourself or others when the primary beam (x-ray on) lights are illuminated
  3. Never hold samples during analysis
  4. Be aware of primary beam indicator lights
  5. Handle and use with respect
  6. Store securely – obey local storage requirements
  7. If you have a Safety Emergency, notify your Radiation Safety Officer (RSO) and analyzer vendor

Gas Chromatography



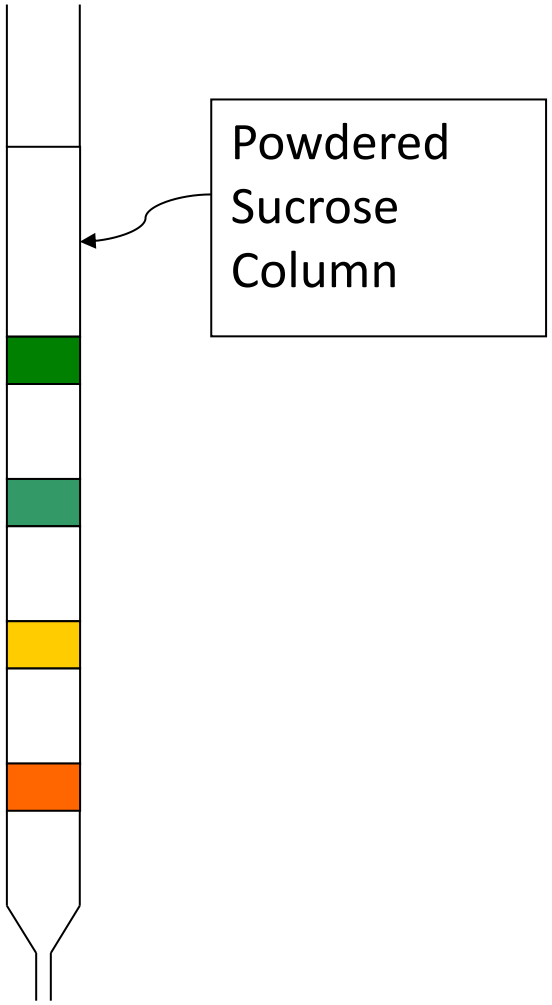
In the animation below the red molecules are more soluble in the liquid (or less volatile) than are the green molecules.



History

- Chromatography was discovered by Michael Tswett in the first part of the 20th century - in 1906.
- The technique was discovered in the course of investigating plant pigments
- Tswett passed pentane solutions of plant pigments through a bed of powdered sucrose and observed the formation of colored bands – hence the name chromatography.
- At the time he speculated that it would be possible to separate colorless compounds by the same technique, but he never did it himself.

Tswett's Original Experiment



Chromatography

- Chromatography is a physical method of separation
- Components to be separated are distributed between two phases
- One phase is stationary while the other phase is mobile which moves through it in a definite direction
- The chromatographic process occurs due to differences in the distribution constant of the individual sample components

Associated Terminology

- **Mobile phase**
  - Moves in a definite direction. Liquid (LC), Gas (GC).
    - The mobile phase moves through the chromatography column (the stationary phase) where the sample interacts with the stationary phase and is separated.

- **Stationary phase - Immobilized phase**
  - Immobilized on the support particles or on the inner wall of the column tubing.
  - Examples : Silica layer - Thin Layer Chromatography

- **Sample (Analyte)**: Substance to be analyzed in chromatographic analysis.

- **Solvent** : Any substance capable of solubilizing another substance.

- **Chromatograph** - equipment that enables a sophisticated separation

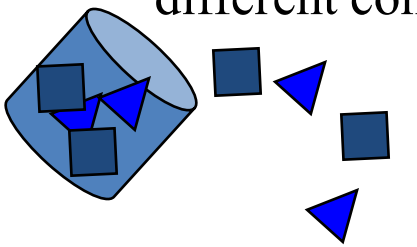
  e.g. Gas chromatography or Liquid chromatography

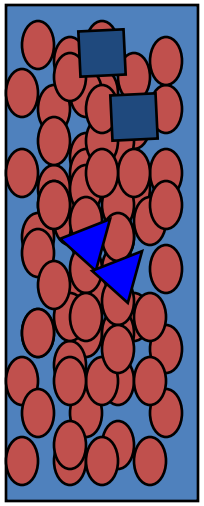
- **Eluent** - Fluid entering column OR solvent that carries the analyte.

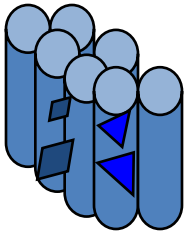
- **Eluate** - Mobile phase leaving the column.

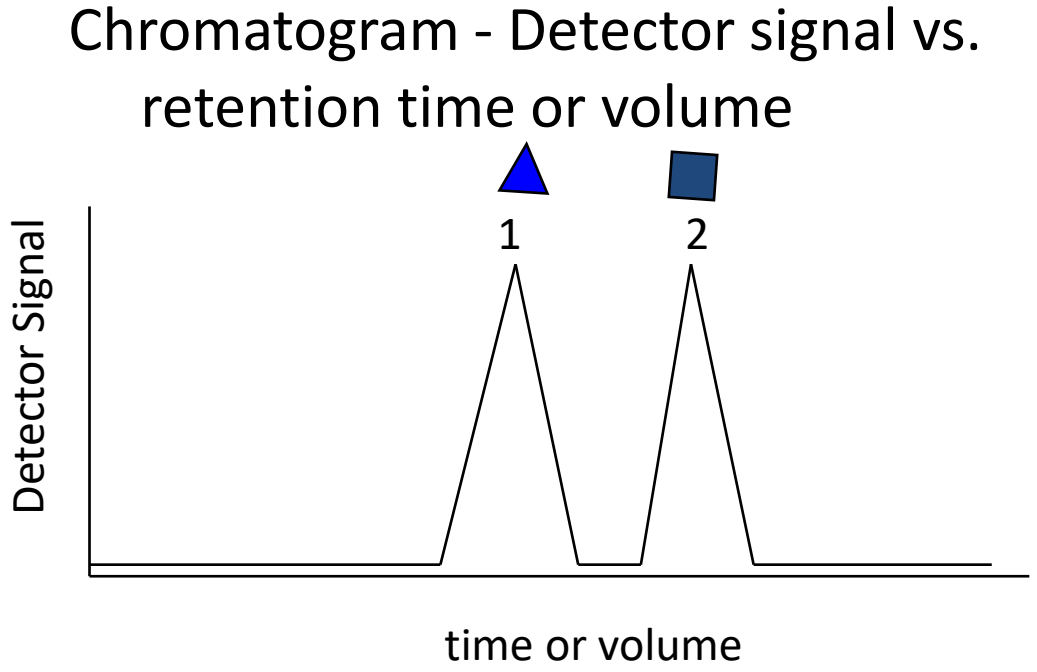
- **Retention time :** Time taken by a particular analyte to pass through the system (from the column inlet to the detector) under set conditions.

- **Chromatogram**
  - Visual output of the chromatograph.
  - Separation - Different peaks or patterns on the chromatogram correspond to different components of the separated mixture.









- X- axis - Retention time
- Y-axis - Signal
- Signal is proportional to the concentration of the specific analyte separated.

Chromatography

- Chromatography, works by allowing the molecules present in the mixture to distribute themselves between a stationary and a mobile medium.
  - Molecules that spend most of their time in the mobile phase are carried along faster.
- Virtually all stable molecules will survive one or more types of chromatographic separation
- Depending upon the system used, the purity of chromatographed compounds can be from modest to very high – even to analytical purity

Chromatography

- The separation of the different components in the mixture is based on
  - Polarity
  - Boiling point
  - Ionic strength
  - Size

Classification of Chromatography

According to the packing of the stationary phase

1. **Planer chromatography (TLC)**
   - **Thin layer / Paper chromatography** The stationary phase is a thin layer supported on glass, plastic or aluminium plates.
   - The stationary phase is a thin film of liquid supported on an inert support

2. **Column chromatography (CC)**
   - The stationary phase is packed in a glass column

According to the mobile phase

- **Liquid chromatography:**

  mobile phase is a liquid (LLC, LSC)

- **Gas chromatography:**

  mobile phase is a gas (GSC, GLC)

Different Types of chromatography

Mode or type	Stationary phase	Mobile phase	Mechanism
Adsorption Chromatography	Solid that attracts the solutes	Liquid or gas	Solutes move at different rates according to the forces of attraction to the stationary phase.
Partition Chromatography	Thin film of liquid formed on the surface of a solid inert support	Liquid or gas	Solutes equilibrate between the 2 phases according to their partition coefficients
Ion Exchange Chromatography	Solid resin that carries fixed ions & mobile counter ions of opposite charge attached by covalent bonds	Liquid containing electrolytes	Solute ions of charge opposite to the fixed ions are attracted to the resin by electrostatic forces & replace the mobile counter ions.
Molecular Exclusion Chromatography	Porous gel with no attractive action on solute molecules	Liquid	Molecules separate according to their size: 1.Smaller molecules enter the pores of the gel, and need a larger volume of eluent. 2.Larger molecules pass through the column at a faster rate.
Affinity Chromatography	Solid on which specific molecules are immobilized	Liquid or gas	Special kind of solute molecules interact with those immobilized on the stationary phase

What is GC ?

- GC is a Separation Technique
- Sample is usually a complex mixture we require to separate into constituent components

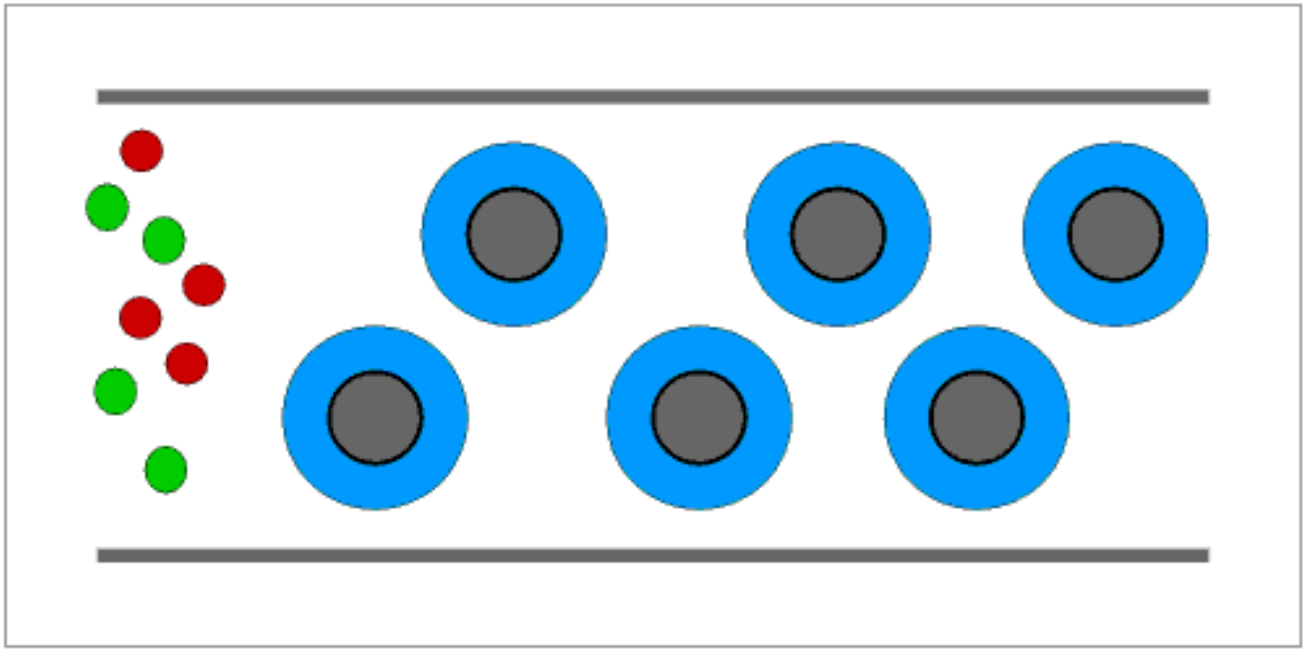
Gas - Solid Chromatography (GSC)

- The stationary phase, in this case, is a solid like silica or alumina
- It is the affinity of solutes towards the stationary phase which determines, in part, the retention time
- The mobile phase is, of course, a suitable carrier gas
- The use of GSC in practice is considered marginal when compared to gas liquid chromatography

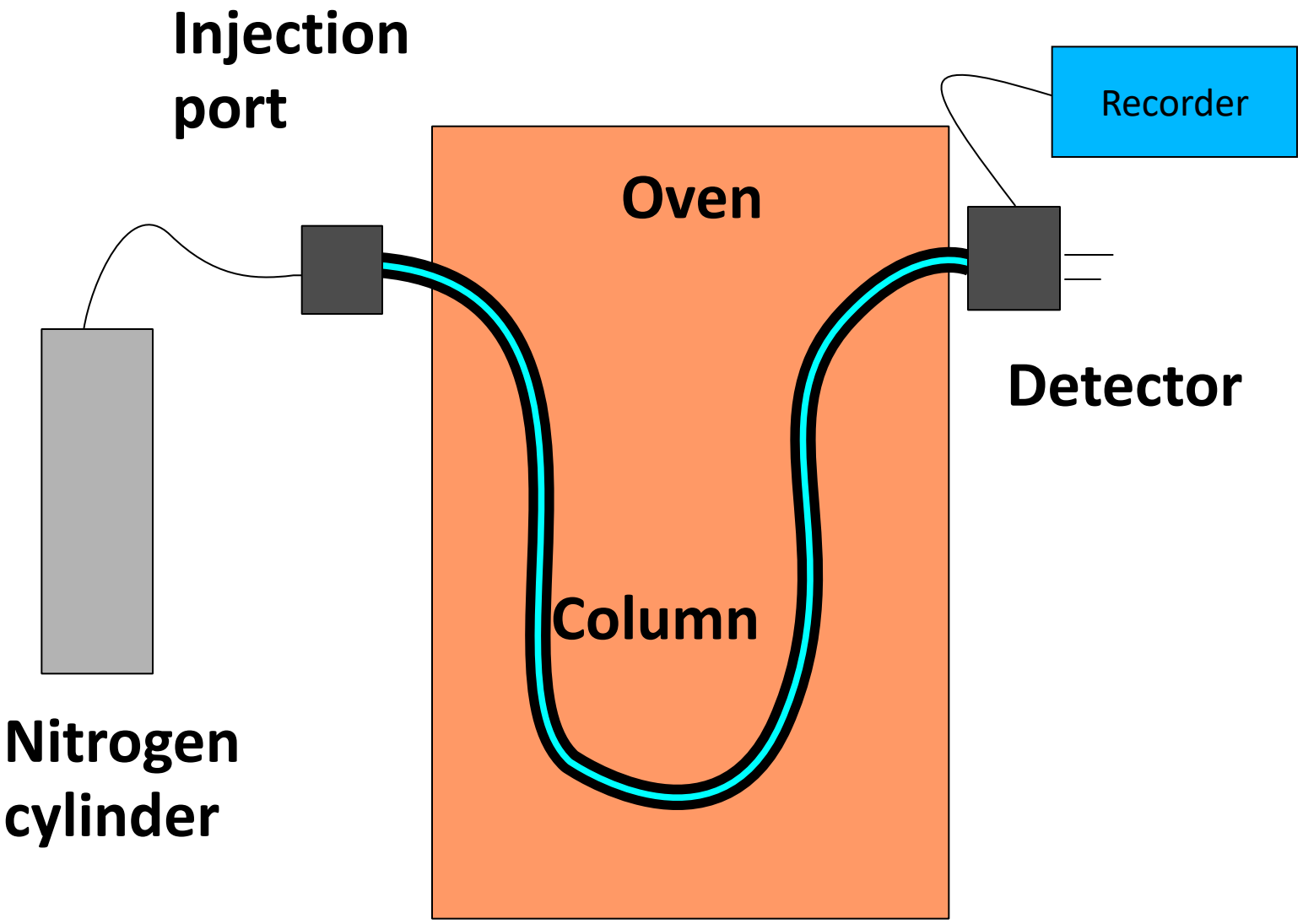
Gas - Liquid Chromatography (GLC)

- Partition of molecules between gas (mobile phase) and liquid (stationary phase).
- The stationary liquid phase with very low volatility is held on particles of a solid support.
- The mobile phase is a suitable carrier gas (Helium, Nitrogen)
- GLC is the most widely used technique for the separation of volatile species
- The wide variety of stationary phases are available
- Column preparation is quite easy
- GLC is simply called GC

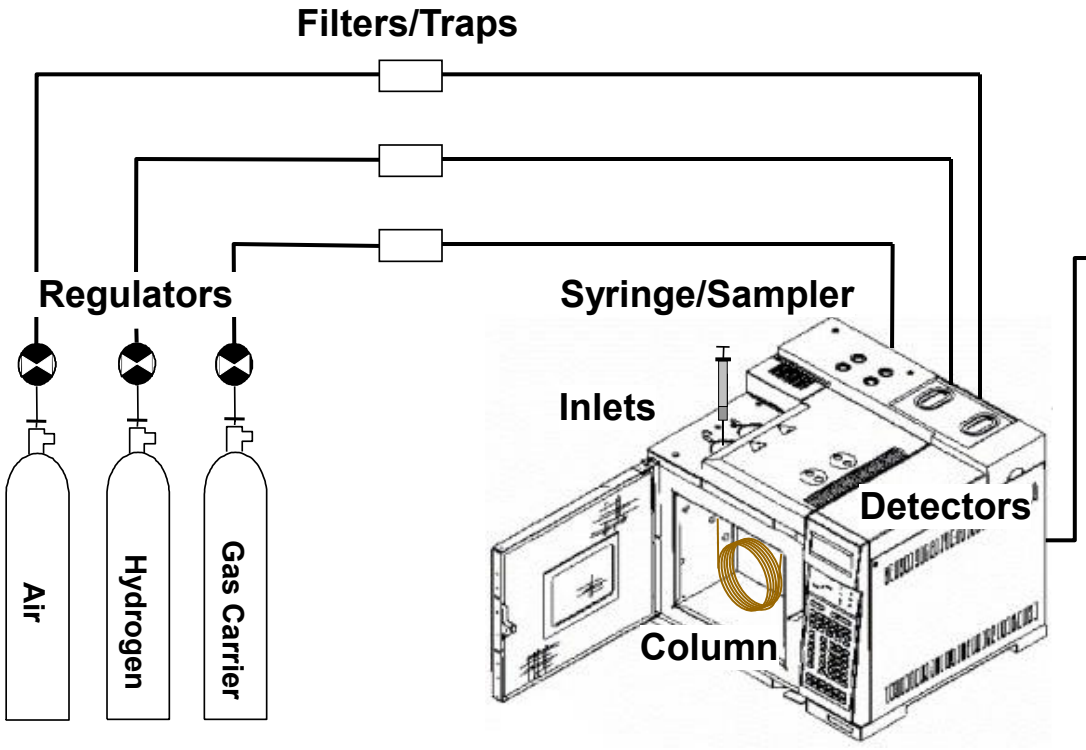
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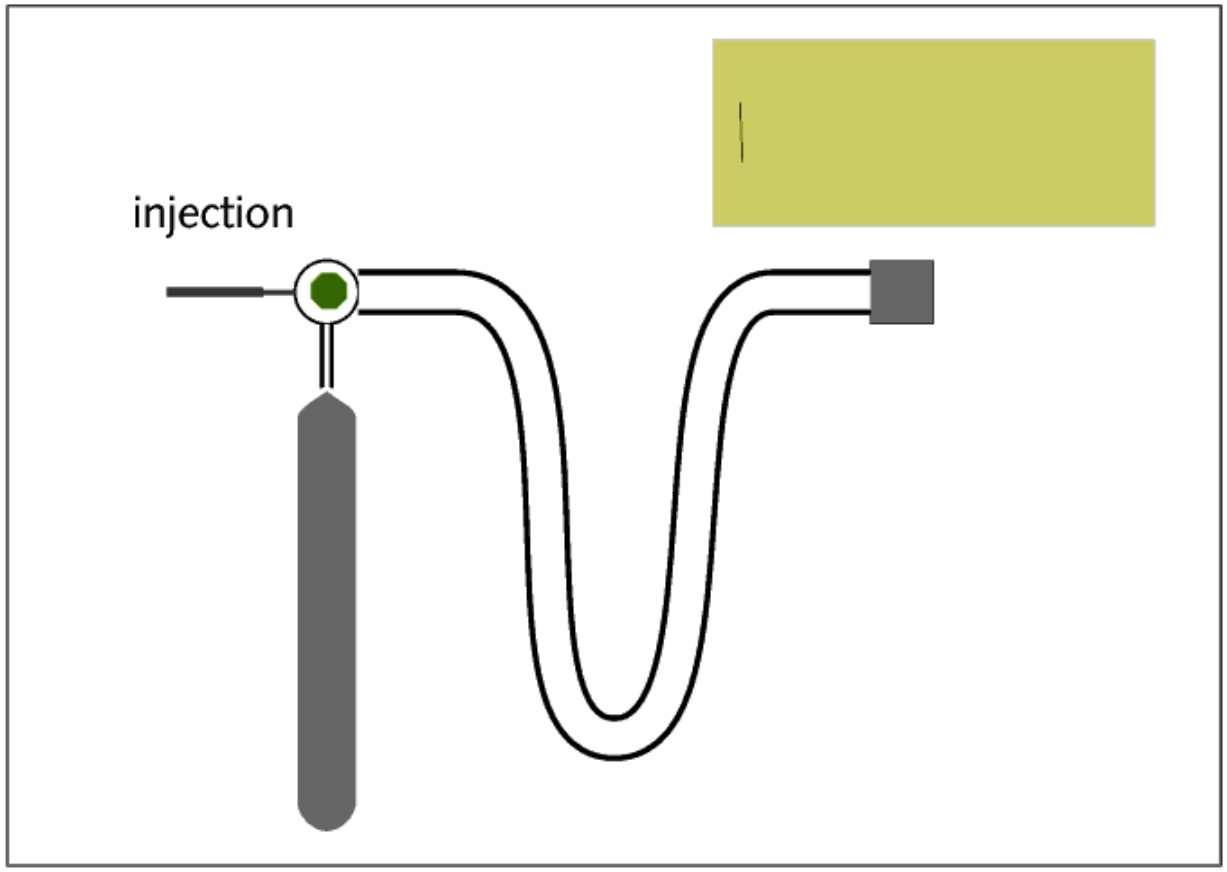
Schematic Diagram of Gas Chromatography



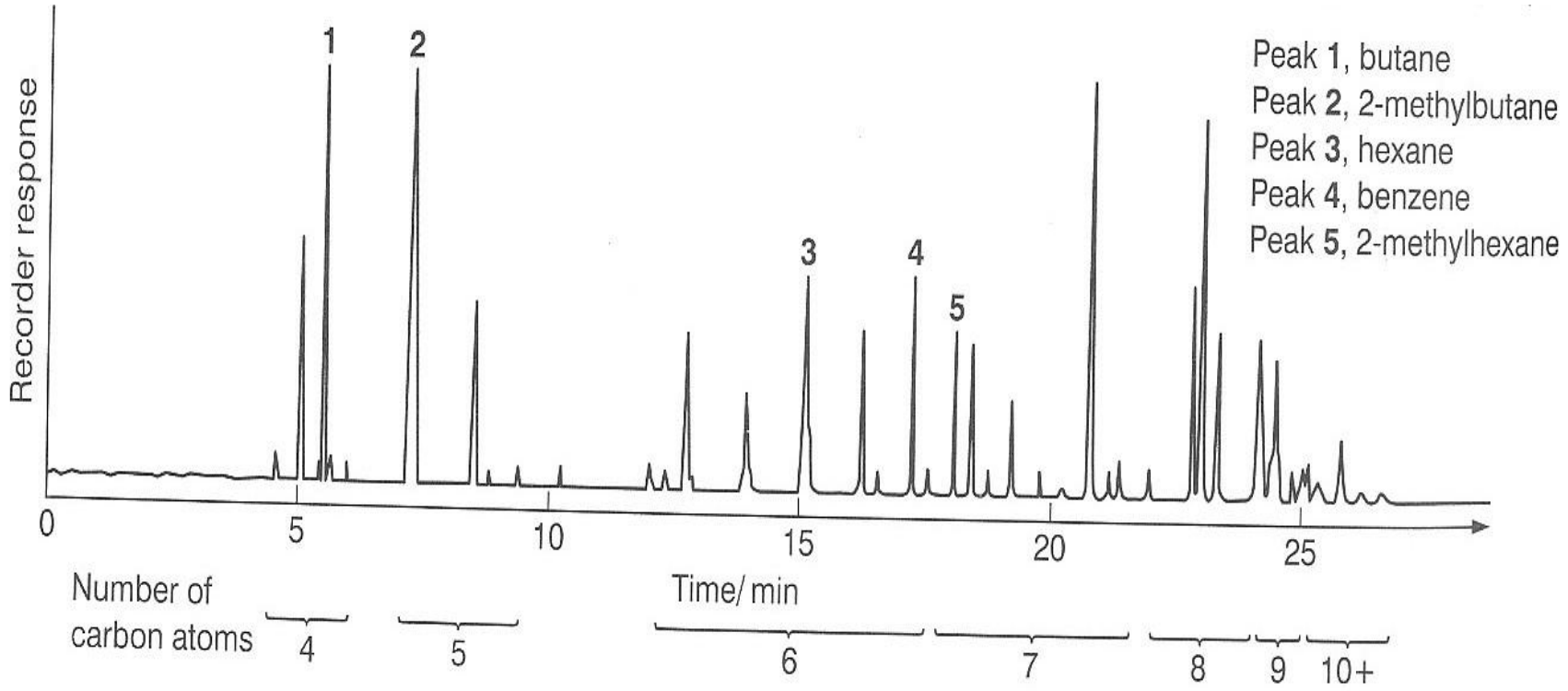
Gas Chromatography



- gas system
- inlet
- column
- detector
- data system


injection

Chromatogram of petrol



Some additional peaks (unlabeled) are due to the presence of other fragments present in the petrol

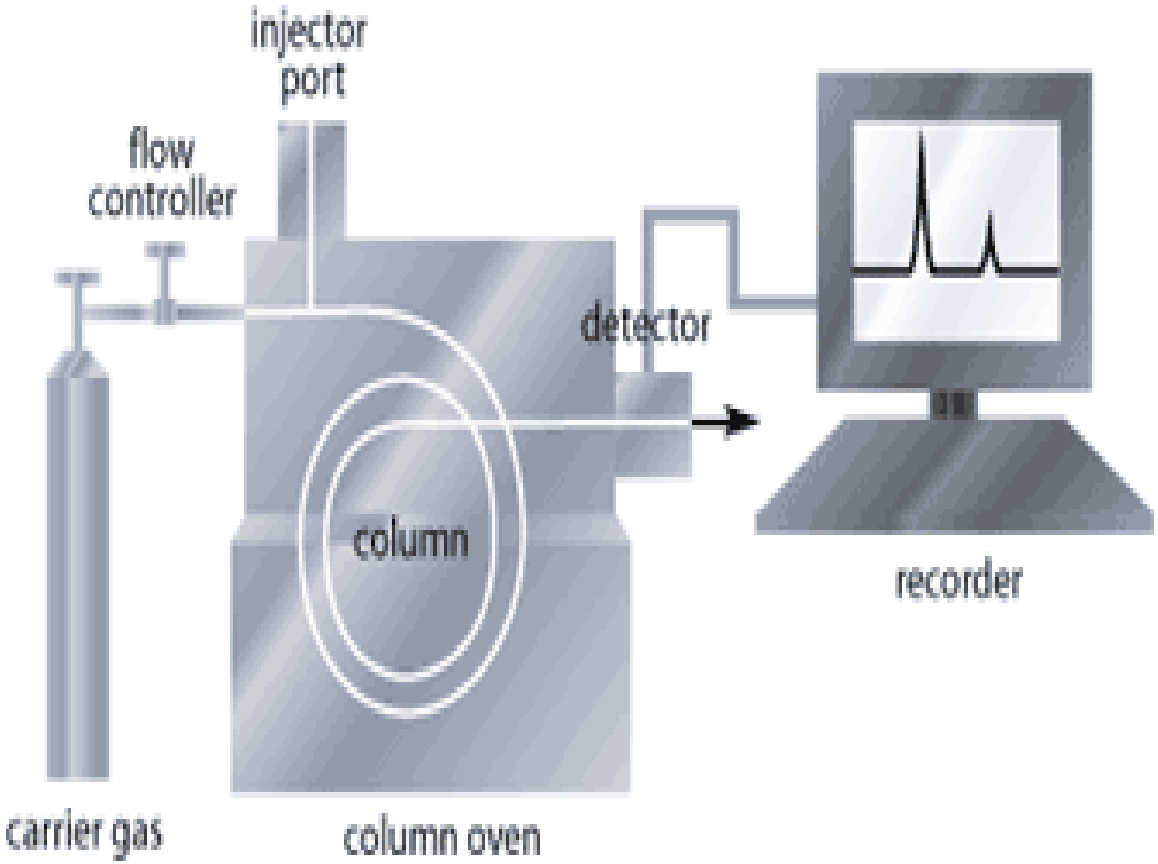
Chromatographic Analysis

The number of components in a sample is determined by the number of peaks.

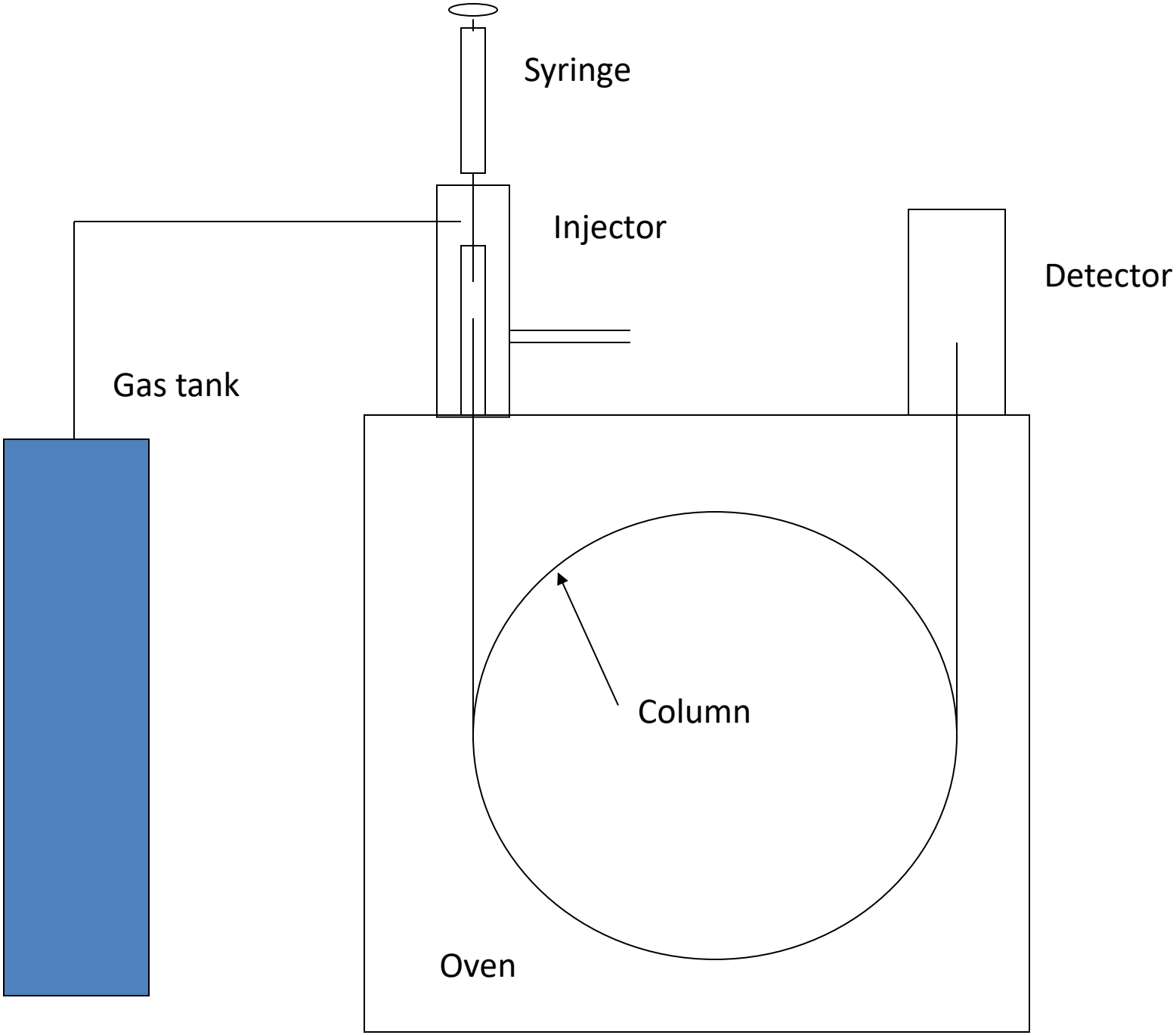
The amount of a given component in a sample is determined by the area under the peaks.

The identity of components can be determined by the given retention times.

GC Instrumentation



1. Carrier gas
2. Flow controller (regulators & Flow meters)
3. Injection devices
4. Column
5. Temperature control devices (column oven)
6. Detector
7. Recorders, Integrators & display unit


Syringe
Injector
Detector
Gas tank
Column
Oven

CARRIER GAS

1. **Hydrogen**
   - Better thermal conductivity
   - Disadvantage: it reacts with unsaturated compounds & inflammable

2. **Helium**
   - Excellent thermal conductivity
   - It is expensive

3. **Nitrogen**
   - Reduced sensitivity
   - It is inexpensive

Ideal Inert Gas characteristics:

- Inertness
- Suitable for the detector
- High purity
- Easily available
- Cheap
- Should not cause the risk of fire
- Should give best column performance

Flow Controller and Injection Device

- Delivers the gas with uniform pressure/flow rate.
- Flow meters:
  1. Rota meter
  2. Soap bubble flow meter

- A GC syringe penetrates a septum to inject sample into the vaporization chamber

- Instant vaporization of the sample, 280 °C

- Carrier gas transports the sample into the head of the column

- Purge valve controls the fraction of sample that enters the column

Temperature Control Devices/Oven

- **Preheaters:**
  - Convert sample into its vapour form, present along with injecting devices

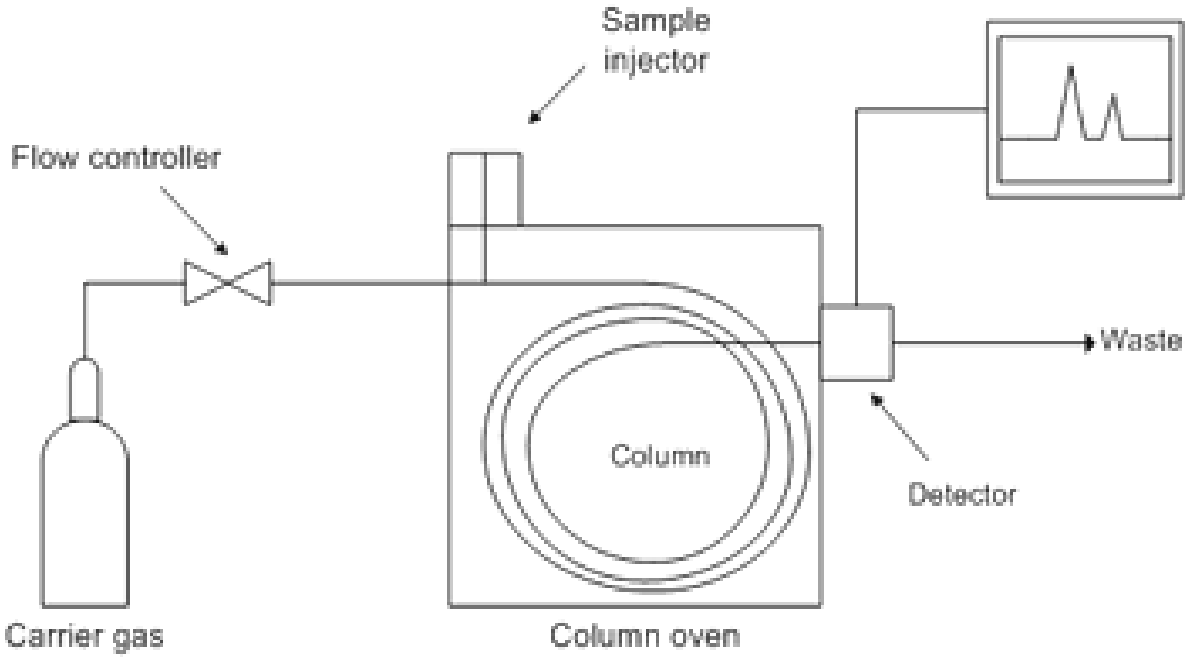
- **Thermostatically controlled oven:**
  - Temperature maintenance in a column is highly essential for efficient separation.

1. **Isothermal Programming**- run at one constant temperature
2. **Temperature programming** - Start at low temperature and gradually ramp to higher temperature

Column

- Most important part of GC is its **COLUMN**
- Made up of **glass** or **stainless steel**
- Glass column- inert , highly fragile

1. **Packed column:**
   - Columns are available in a packed manner
   - S.P for GLC: polyethylene glycol, esters, amides, hydrocarbons, polysiloxanes
2. **Open tubular or Capillary column or Golay column:**
   - Long capillary tubing 30-90 M in length
   - Uniform & narrow d.m of 0.025 - 0.075 cm
   - Made up of stainless steel & form of a coil
   - Disadvantage: more sample cannot loaded
3. **SCOT columns (Support coated open tubular column**
   - Improved version of Golay / Capillary columns, have small sample capacity
   - Made by depositing a micron size porous layer of supporting material on the inner wall of the capillary column
   - Then coated with a thin film of liquid phase


Sample
injector
Flow controller
Waste
Column
Detector
Carrier gas
Column oven



Most Common Stationary Phases

- Separation of mixture of polar compounds Carbowax 20M (polyethylene glycol)
- Separation of mixtures of non-polar compounds OV101 or SE-30 (polymer of methylsilicone)
- Methylester of fatty acids DEGS (diethylene glycol succinate)

Fused Silica Open Tubular (FSOT) Columns

- Coated with polymer, crosslinked
  - Polydimethyl siloxane (non-polar)
  - Poly(phenylmethyldimethyl) siloxane (10% phenyl)
  - Poly(phenylmethyl) siloxane (50% phenyl)
  - Polyethylene glycol (polar)
  - Poly(dicyanoallyldimethyl) siloxane
  - Ploy(trifluoropropyldimethyl) siloxane

Polar vs Non-polar

- Separation is based on the vapor pressure and polarity of the components.
- Within a homologous series (alkanes, alcohols, olefins, fatty acids) retention time increases with chain length (or molecular weight)
- Polar columns retain polar compounds to a greater extent than non-polar
  - C18 saturated vs. C18 saturated methyl ester

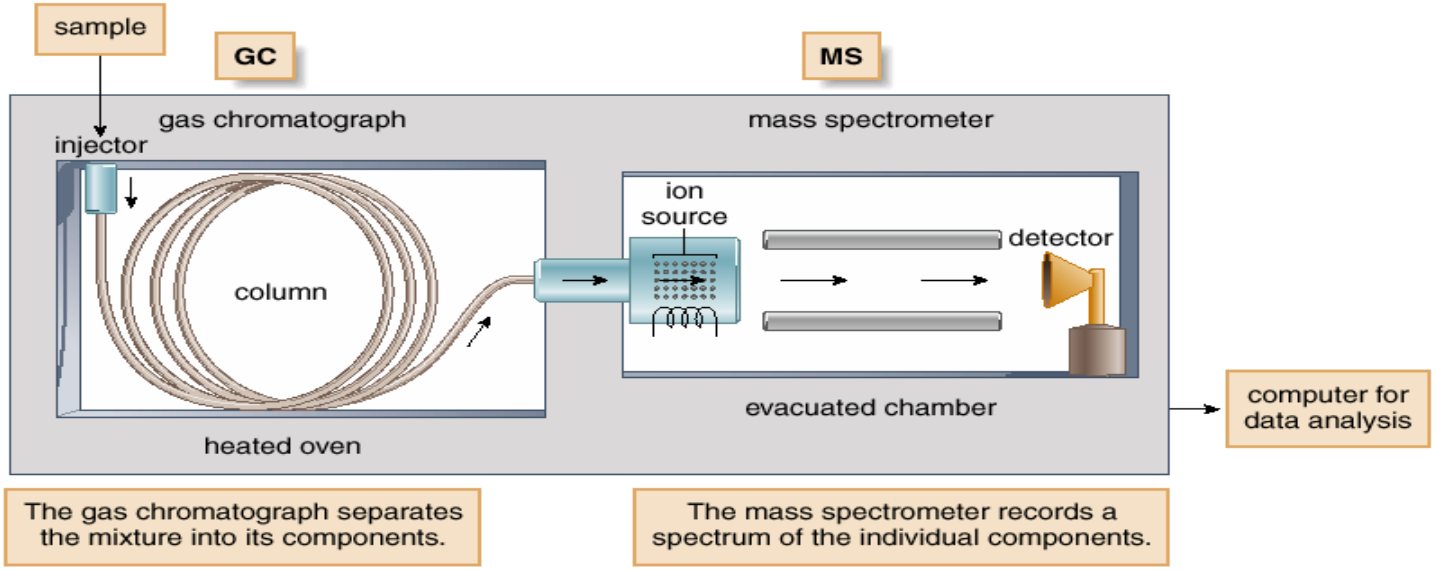
GC Detectors

- Requirements for an ideal detectors-
  - Applicability to wide range of samples
  - Rapidity
  - High sensitivity
  - Linearity
  - Response should be unaffected by temperature, flow rate
  - Non destructive
  - Simple & inexpensive

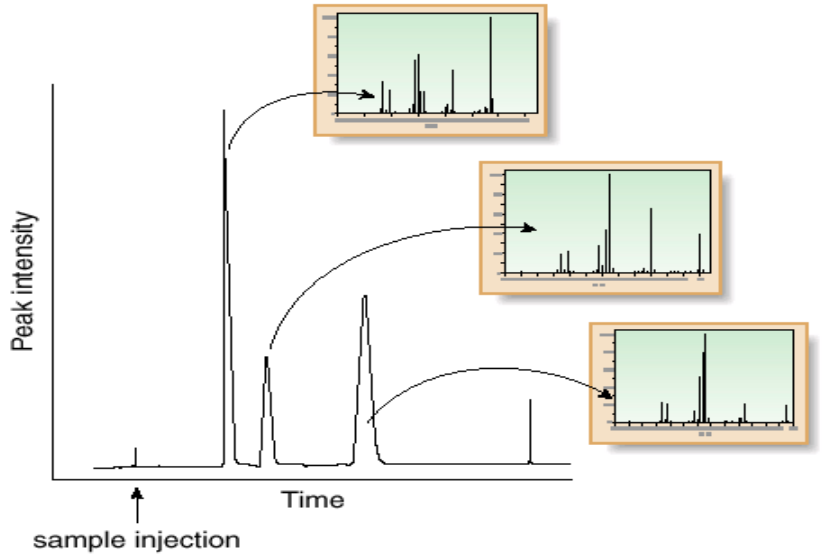
TYPES OF DETECTORS USED WITH GC

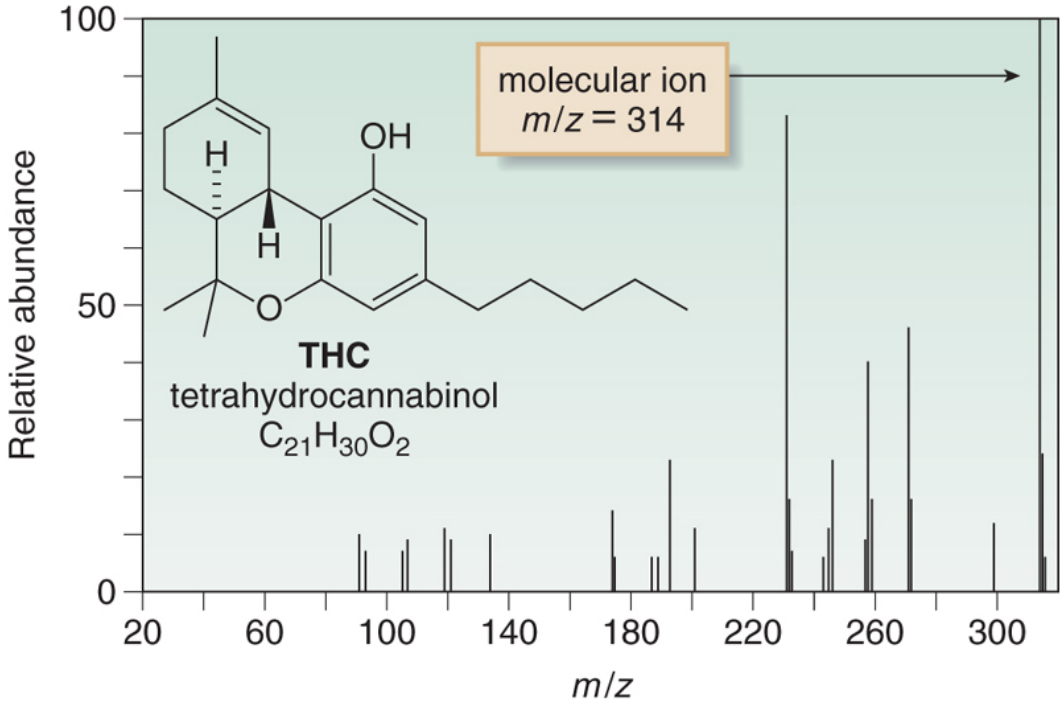
1. Thermal Conductivity Detector **(TCD)**
2. Flame Ionization Detector **(FID)**
3. Argon Ionization Detector **(AID)**
4. Electron Capture Detector **(ECD)**
5. Mass Spectrometry **(MS)**

a. Schematic of a GC–MS instrument



b. GC trace of a three-component mixture. The mass spectrometer gives a spectrum for each component.




100
50
0
Relative abundance
molecular ion
m/z = 314
OH
H
H
O
THC
tetrahydrocannabinol
C21H30O2
20
60
100
140
180
220
260
300
m/z

Applications of GC

- Quantitative & Qualitative analysis of low polarity compounds
- Analytical chemistry, Biochemistry, Petrochemical, Environmental monitoring
- Measure picomoles of a substance in a 1 ml liquid sample, or parts-per-billion concentrations in gaseous samples
- Measuring toxic substances in soil, air or water.

Forensic applications:

1. **Environmental Forensics :** Oraganic Pollutants
2. **Criminal Forensics :** Analyze the particles (Fibre) from a human body in order to help link a criminal to a crime.
3. **Law Enforcement :** Detection of illegal narcotics,
4. **Forensic Toxicology :** Find drugs and/or poisons in biological specimens of suspects, victims, or the deceased.
5. **Sports Anti-Doping Analysis :** Test athletes' urine samples
6. **Explosive Detection:** at airports, stations and other significant public and government buildings etc.
7. **Forensic Chemistry:** from spoilage or Adultration - aromatic compounds, esters, fatty acids, alcohols, aldehydes, terpenes
8. **Consumer Forensics:** Food, beverage and perfume

Advantages and Disadvantages of GC

Advantages

- Fast analysis
  - Typically minutes (even sec.)
- High Resolution
  - Record N~1.3 x 10^6
- Sensitive detectors (easy ppm, often ppb)
- Highly accurate quantification (1-5 % RSD)
- Automated systems
- Non-destructive*
  - Allows online coupling to Mass Spec.
- Small sample (mL)
- Reliable and relatively simple
- Low cost

Disadvantages

- Limited to volatile samples

  T limited to ~ 380 °C
  Need P_{vap} ~ 60 Torr at that temperature

- Not suitable for thermally labile samples
- Some samples may require extensive preparation
- Requires spectroscopy (usually MS) to confirm peak identify